7/8/2026

Review: • density matrix - describes both pure & mixed states

$$\hat{\rho} = \sum_n p_n |\psi_n\rangle\langle\psi_n|$$

• $Tr(\hat{\rho}^2) \equiv$ purity $= 1$ for pure states

< 1 for mixed states

$= \frac{1}{d}$ for maximally mixed states

• $\langle \hat{A} \rangle = Tr(\hat{\rho}\hat{A})$ average value of observable $\hat{A}$

• Entangled states cannot be written as product state

• Schmidt decomposition basis where each state $|x_n\rangle$ of A is uniquely associated w/ $|\gamma_n\rangle$ of B



$$|\psi\rangle = \sum_n a_n |x_n\rangle_A |\gamma_n\rangle_B$$

$$\hat{\rho}^A = \sum |a_n|^2 |x_n\rangle\langle x_n|$$
$$\hat{\rho}^B = \sum |a_n|^2 |\gamma_n\rangle\langle\gamma_n|$$

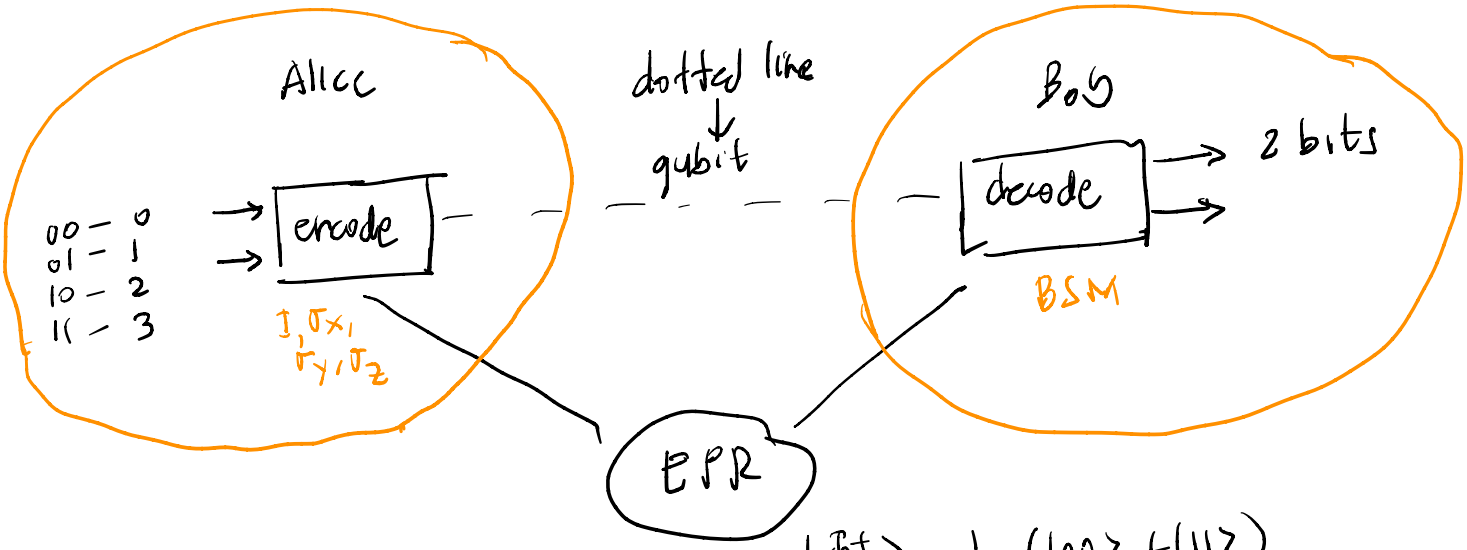
} reduced density matrices are mixed if $|\psi\rangle$ is entangled

• (MES) maximally entangled state if a_n's are equal

• Bell states - MES in qubit Hilbert space

Dense Coding (Bennett & Weisner) 1992

- Use EPR pair + simple gates to transmit 2 bits of information



$$|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$$

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

'dense' because w/ 1 qubit you get 2 bits

1. Share EPR
2. Alice depending on whether she wants to send 0, 1, 2, 3 does I, σ_x, σ_y, σ_z
3. Alice sends the qubit to Bob
4. Bob does a CNOT on the qubits, Alice's qubit is control, does H on Alice's qubit
5. Bob measures both qubits

① Share EPR

② Alice depending on whether she wants to send 0, 1, 2, 3 does $I, \sigma_x, \sigma_y, \sigma_z$

③ Alice sends the qubit to Bob

④ Bob does a CNOT on the qubits, Alice's qubit is control, does H on Alice's qubit

⑤ Bob measures both qubits

$$|\beta^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$$

Alice's side

0. $|\psi_0\rangle = (I \otimes I)|\beta^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$

1. $|\psi_1\rangle = (\sigma_x \otimes I)|\beta^+\rangle = \frac{1}{\sqrt{2}}(|10\rangle + |01\rangle)$

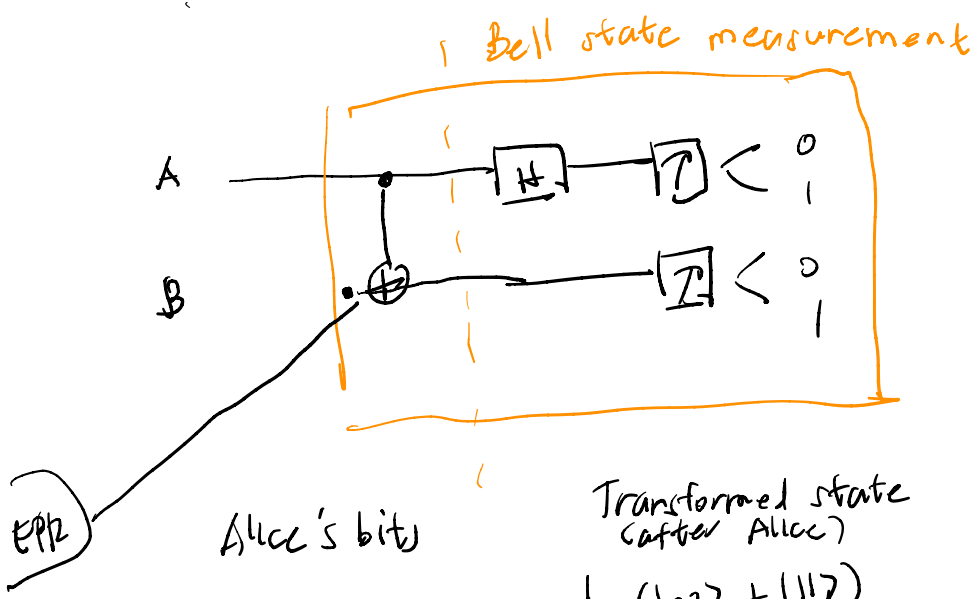
2. $|\psi_2\rangle = (\sigma_z \otimes I)|\beta^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle - |11\rangle)$

3. $|\psi_3\rangle = (\sigma_y \otimes I)|\beta^+\rangle = \frac{1}{\sqrt{2}}(i|10\rangle - i|01\rangle)$ $(-i)$

$\sigma_y|0\rangle = i|1\rangle$

$\sigma_y|1\rangle = -i|0\rangle$

$= \frac{1}{\sqrt{2}}(-|10\rangle + |01\rangle$



Alice's bits	Transformed state (after Alice)	After CNOT
0	$\frac{1}{\sqrt{2}}(\|00\rangle + \|11\rangle)$	$\frac{1}{\sqrt{2}}(\|00\rangle + \|10\rangle)$
1	$\frac{1}{\sqrt{2}}(\|10\rangle + \|01\rangle)$	$\frac{1}{\sqrt{2}}(\|11\rangle + \|01\rangle)$
2	$\frac{1}{\sqrt{2}}(\|00\rangle - \|11\rangle)$	$\frac{1}{\sqrt{2}}(\|00\rangle - \|10\rangle)$
3	$\frac{1}{\sqrt{2}}(-\|10\rangle + \|01\rangle)$	$\frac{1}{\sqrt{2}}(-\|11\rangle + \|01\rangle)$

↓ factor!

0	$\frac{1}{\sqrt{2}}(\|0\rangle + \|1\rangle)\|0\rangle$
1	$\frac{1}{\sqrt{2}}(\|1\rangle + \|0\rangle)\|1\rangle$
2	$\frac{1}{\sqrt{2}}(\|0\rangle - \|1\rangle)\|0\rangle$
3	$\frac{1}{\sqrt{2}}(-\|1\rangle + \|0\rangle)\|1\rangle$

↓

Do Hadamard on Alice's qubit

$\|0\,0\rangle$
$\|0\,1\rangle$
$\|1\,0\rangle$
$\|1\,1\rangle$

Measure on computational basis

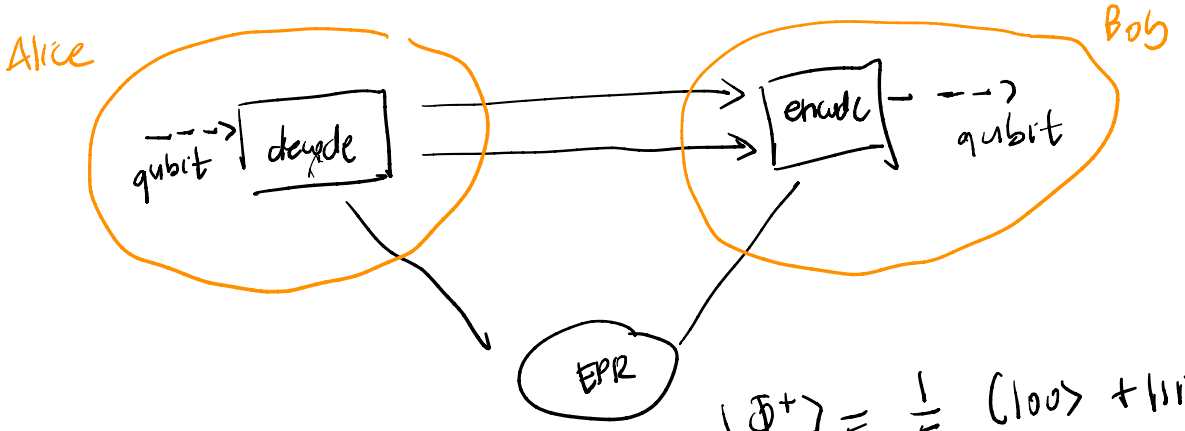
$H: |0\rangle \to |+\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}}$

$|1\rangle \to |-\rangle = \frac{|0\rangle - |1\rangle}{\sqrt{2}}$

Quantum Teleportation

Bennett, 1993

- Transmit classical information to recreate a quantum state



$$|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$$

Alice's qubit $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$

EPR pair $|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$

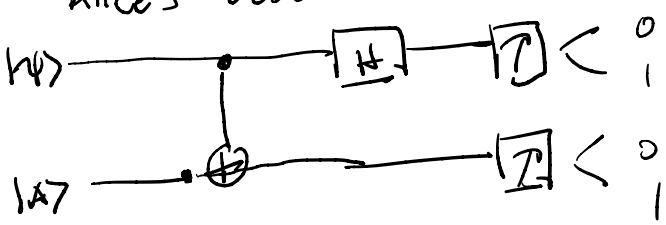
3-qubit state

$$|\psi\rangle \otimes |\Phi^+\rangle = \frac{1}{\sqrt{2}}\left(\alpha|0\rangle \otimes (|00\rangle + |11\rangle) + \beta|1\rangle \otimes (|00\rangle + |11\rangle)\right)$$

$$= \frac{1}{\sqrt{2}}(\alpha|000\rangle + \alpha|011\rangle + \beta|100\rangle + \beta|111\rangle)$$

(AAB)

Alice's 'decode'



$H|0\rangle = |+\rangle$

$H|1\rangle = |-\rangle$

$$(H \otimes I \otimes I)\ (CNOT \otimes I)\ |\psi\rangle \otimes |\Phi^+\rangle$$

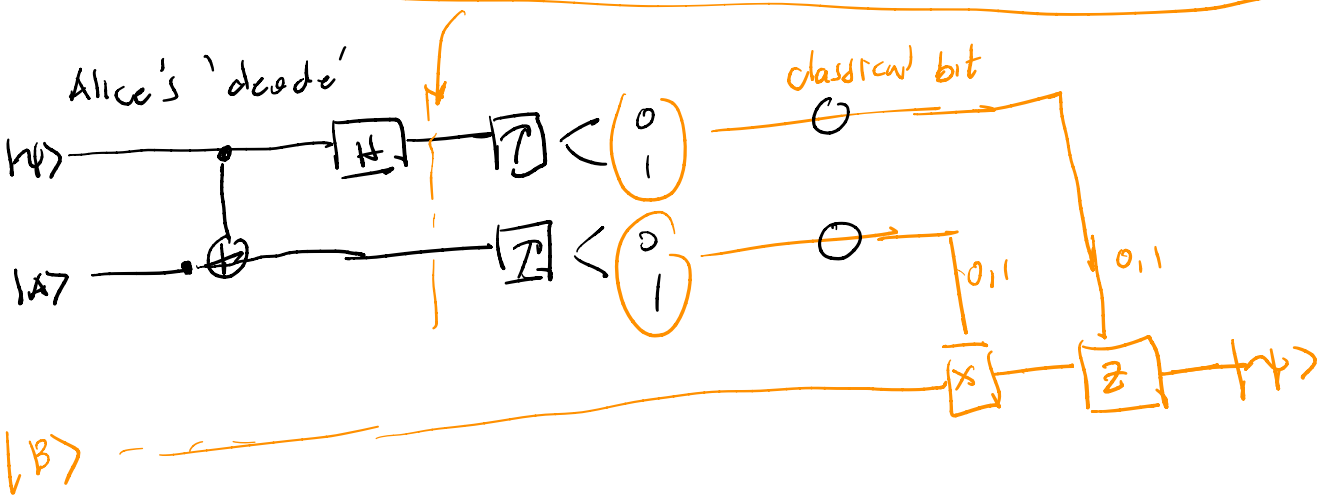
$$= (H \otimes I \otimes I)\ \frac{1}{\sqrt{2}}(\alpha|000\rangle + \alpha|011\rangle + \beta|110\rangle + \beta|101\rangle)$$

$$= \frac{1}{2}(\alpha|000\rangle + \alpha|100\rangle + \alpha|011\rangle + \alpha|111\rangle + \beta|010\rangle - \beta|110\rangle + \beta|001\rangle - \beta|101\rangle)$$

$$\frac{1}{2}\left(\alpha|000\rangle + \alpha|100\rangle + \alpha|011\rangle + \alpha|111\rangle + \beta|010\rangle - \beta|110\rangle + \beta|001\rangle - \beta|101\rangle\right)$$

factor!

$$\frac{1}{2}\left(|00\rangle(\alpha|0\rangle+\beta|1\rangle) + |01\rangle(\alpha|1\rangle+\beta|0\rangle) + |10\rangle(\alpha|0\rangle-\beta|1\rangle) + |11\rangle(\alpha|1\rangle-\beta|0\rangle)\right)$$



Bob receives

00

01

10

11

Bob's qubit

$\alpha|0\rangle + \beta|1\rangle$

$\alpha|1\rangle + \beta|0\rangle$

$\alpha|0\rangle - \beta|1\rangle$

$\alpha|1\rangle - \beta|0\rangle$

'encode'

I $\quad \alpha|0\rangle + \beta|1\rangle$

σ_x $\quad \alpha|0\rangle + \beta|1\rangle$

σ_z $\quad \alpha|0\rangle + \beta|1\rangle$

σ_y $\quad -i(\alpha|0\rangle + \beta|1\rangle)$

$\sigma_x\sigma_z$